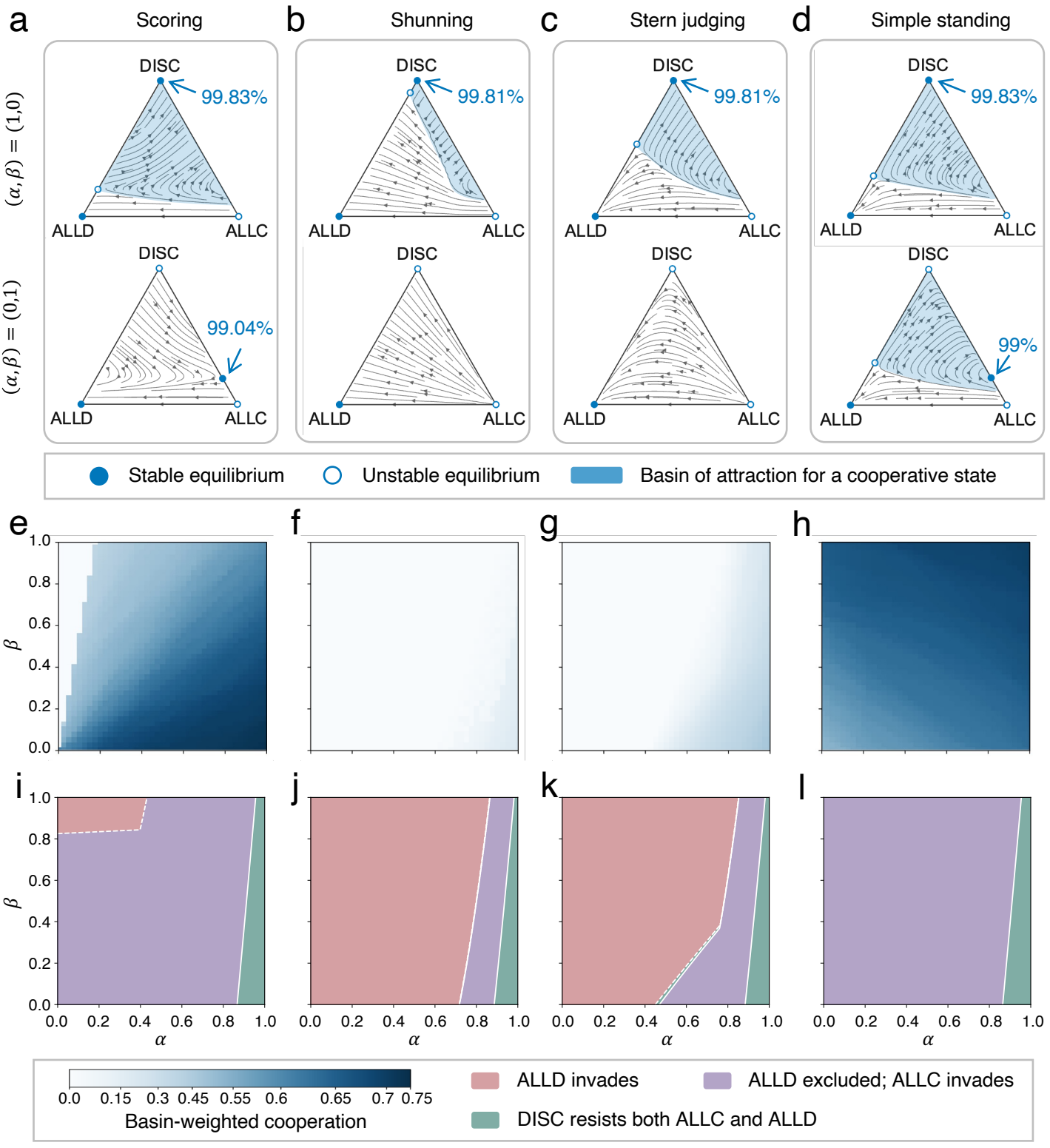

a
Scoring
b
Shunning
c
Stern judging
d
Simple standing
$(\alpha, \beta) = (1,0)$
$(\alpha, \beta) = (0,1)$
DISC
ALLD
ALLC
99.83%
99.81%
99.81%
99.83%
99.04%
99%
Stable equilibrium
Unstable equilibrium
Basin of attraction for a cooperative state
e
f
g
h
i
j
k
l
β
α
0.0
0.2
0.4
0.6
0.8
1.0
0.0 0.15 0.3 0.45 0.55 0.6 0.65 0.7 0.75
Basin-weighted cooperation
ALLD invades
ALLD excluded; ALLC invades
DISC resists both ALLC and ALLD